

CART for Regression (Regression Trees)

↳ Focus on minimizing the residual sum of squares (RSS) or the mean square error (MSE) when making splits

1. Calculate the mean value: For each leaf node, the predicted value is the mean of the target values of the instances that reach that node.
2. Evaluate Potential Splits: For each feature, determine the potential splits that minimize the MSE.
3. Select the Best split: Choose the split with min. MSE.
4. Repeat Recursively: Continue the process until the stopping criteria.

eg: Consider a simple dataset containing information about houses and their prices based on two features: Size and Number of Rooms

x_1 (num) x_2 (cat) y (continuous)

House ID	Size (sq ft)	Rooms	Price (in \$)
1	1500	3	300000 → 30 -
2	2000	4	400000 → 40 .
3	2500	4	450000 → 45 .
4	→ 1800	3	350000 → 35 -
5	2200	4	410000 → 41 .
6	1300	2	250000 → 25 -
7	1700	3	320000 → 32 -
8	1600	3	310000 → 31 -

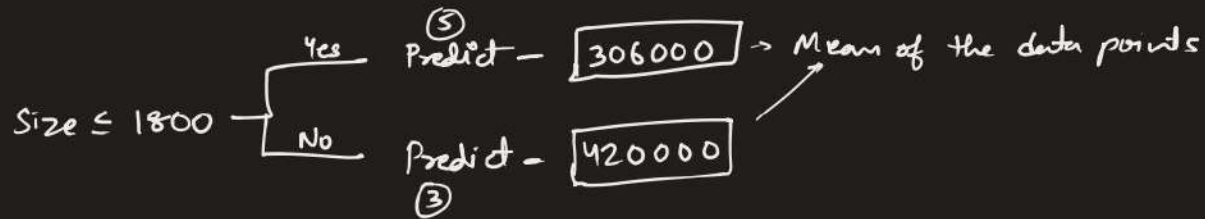
(25)

0

Size:	Threshold	MSE ≤ Thr	MSE > Thr	Weighted MSE
<u>1300</u>	1300	0	28.49	$0 \times \frac{1}{8} + 28.49 \times \frac{7}{8} = 24.93$
1500	1500	...	...	
1600	1600	...	...	
1700	1700	...	...	
1800	1800	10.64	4.67	$\frac{5}{8} \times 10.64 + \frac{3}{8} \times 4.67 = 8.4$
2000	2000	...	...	
2200	2200	...	...	

Assuming $MSE(D_{size \leq 1800})$ is minimum

Stopping Criteria: ($Depth_{max} = 1$)

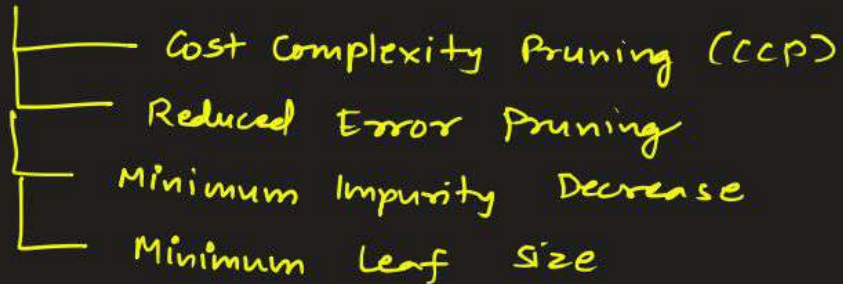


=x=

Pruning of Decision Trees

Overfitting happens when a model is too complex, capturing the noise in the training data rather than the underlying patterns.

Pruning simplifies the model by removing parts of the tree that do not provide significant predictive power.



Cost Complexity Pruning (weakest link pruning)

↳ balance the trade-off between the tree's complexity and its performance on training data.

$$R_{\alpha}(T) = R(T) + \alpha \cdot |T|$$

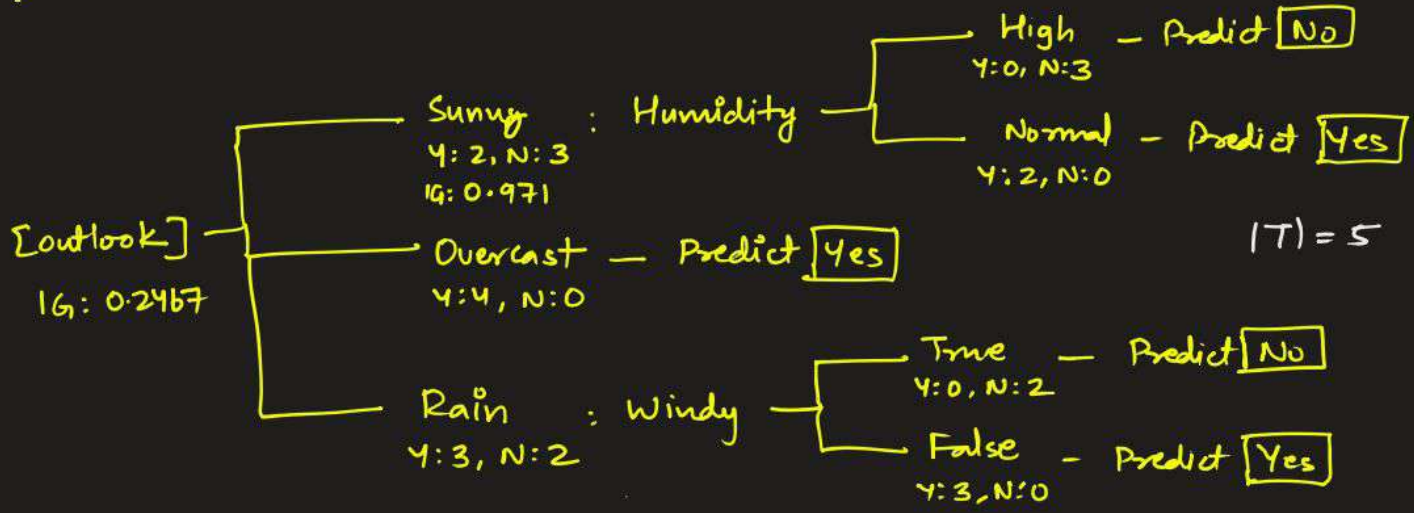
$R(T)$: Total misclassification rate (cost) of the tree

$|T|$: Number of leaf nodes in the tree

α : non-negative parameter that penalizes the # terminal nodes

$R_{\alpha}(T)$: Updated misclassification rate.

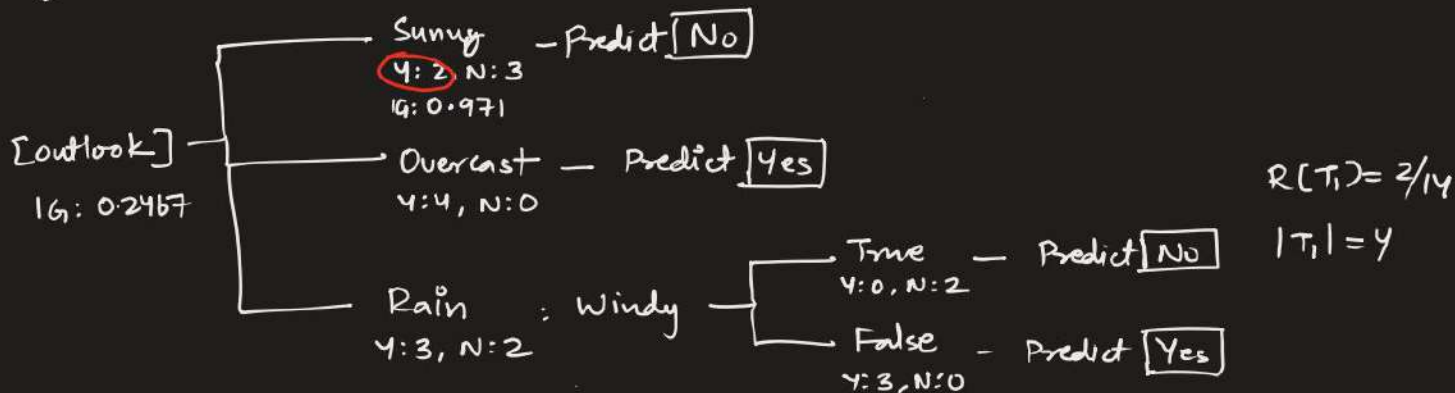
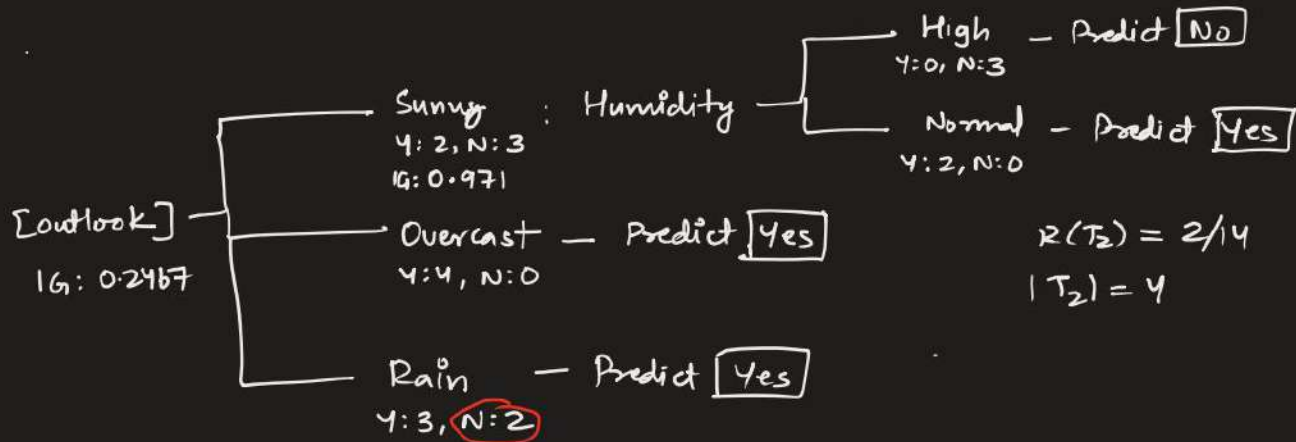
9.

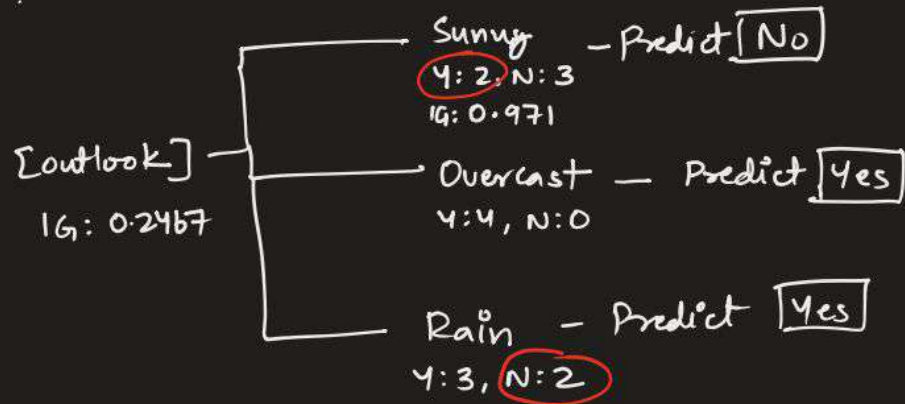


Outlook	Temperature	Humidity	Windy	Play Tennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rain	Mild	High	False	Yes
Rain	Cool	Normal	False	Yes
Rain	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rain	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rain	Mild	High	True	No

$\alpha = 0$

$R_{\alpha}(T) = 0/4 + 0.5 = 0$

T_1  T_2 

T_3 

$$R(T_3) = 4/14$$

$$|T_3| = 3$$

$$T_4; \text{ Predict } - \boxed{\text{Yes}} \quad |T_4| = 1, R(T_4) = 5/14$$

$$\alpha = \underline{\underline{0.5}} \text{ (under-fitting)}$$

$$R_\alpha(T) = 0/14 + 0.5 \times 5 = 2.5$$

$$R_\alpha(T_1) = 2/14 + 0.5 \times 4 = 2.14$$

$$R_\alpha(T_2) = 2/14 + 0.5 \times 4 = 2.14$$

$$R_\alpha(T_3) = 4/14 + 0.5 \times 3 = 1.78$$

$$\underline{\underline{R_\alpha(T_4) = 5/14 + 0.5 \times 1 = 0.86}}$$

↳ Min. Error

$$\underline{\alpha = 0.1}$$

$$R_{\alpha}(T) = 0/14 + 0.1 \times 5 = 0.5$$

$$R_{\alpha}(T_1) = 2/14 + 0.1 \times 4 = 0.54$$

$$R_{\alpha}(T_2) = 2/14 + 0.1 \times 4 = 0.54$$

$$R_{\alpha}(T_3) = 4/14 + 0.1 \times 3 = 0.58$$

$$R_{\alpha}(T_4) = 5/14 + 0.1 \times 1 = 0.457$$

min error

$$\underline{\alpha = 0.02}$$

Min. Error

$$\underline{R_{\alpha}(T)} = 0/14 + 0.02 \times 5 = 0.1$$

$$R_{\alpha}(T_1) = 2/14 + 0.02 \times 4 = 0.22$$

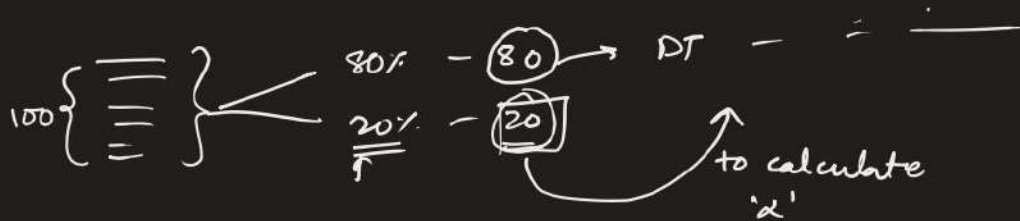
$$R_{\alpha}(T_2) = 2/14 + 0.02 \times 4 = 0.22$$

$$R_{\alpha}(T_3) = 4/14 + 0.02 \times 3 = 0.345$$

$$R_{\alpha}(T_4) = 5/14 + 0.02 \times 1 = 0.37$$

How do we select α ?

We vary the value of α and calculate the missclassification rate on a separate "validation set" which is not used in training and choose the value of α which gives best result on unseen data

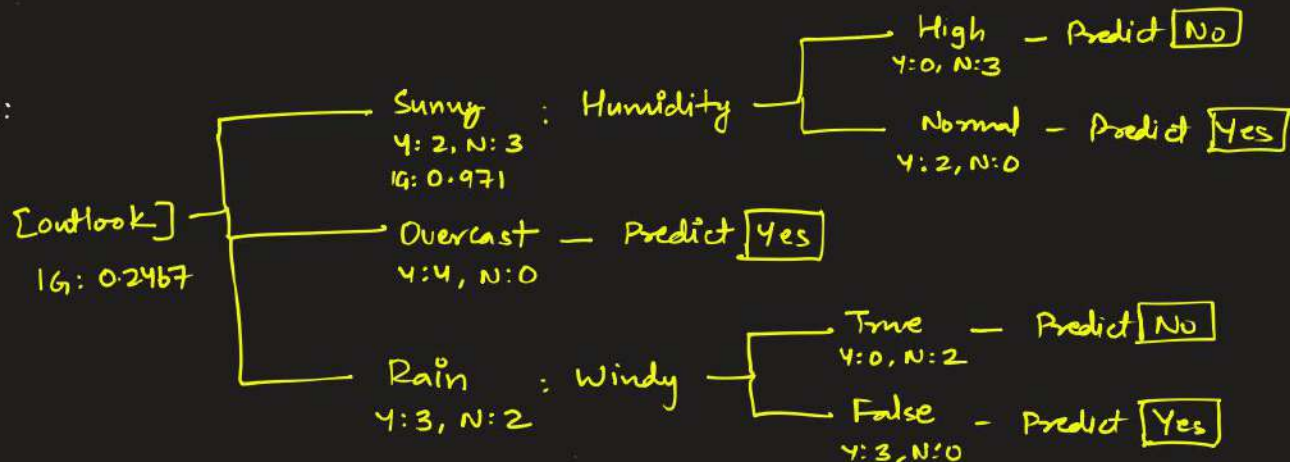


Reduced Error Pruning

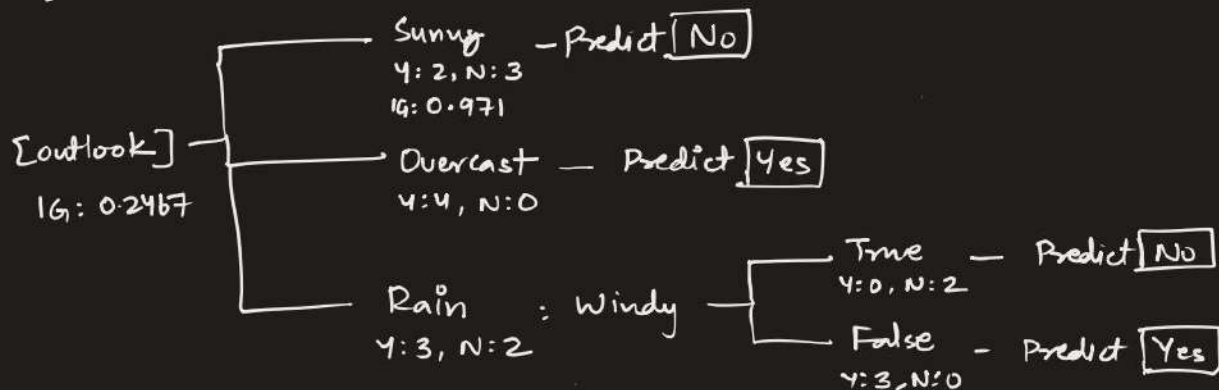
- > Start with a fully grown tree
- > For each node, consider removing it and replacing it with a leaf node that predicts the majority class of the training instances that reach the node
- > Measure the error on the validation set for each constructed tree.
- > Pick the tree with minimum error on validation set.

eg.
#

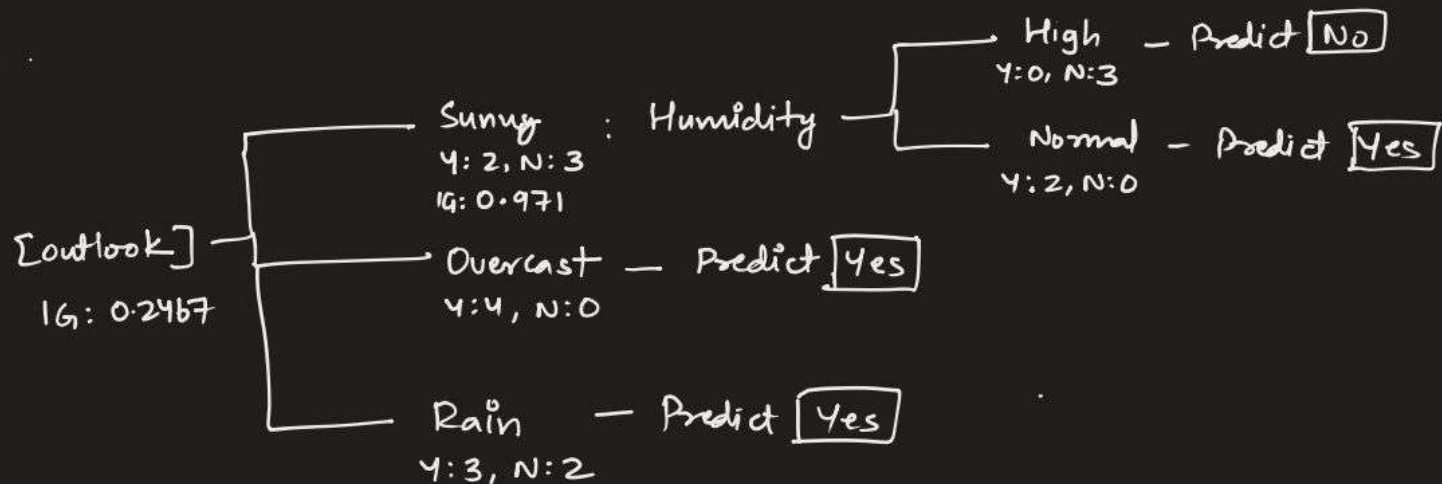
T_0 :



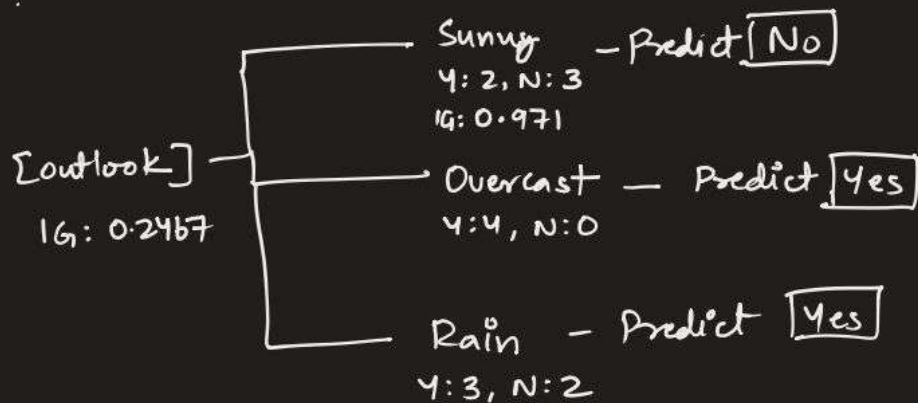
T_1 :



T_2 :



T_3



T_4 : Predict - **Yes**

Validation Set:

Outlook	Temperature	Humidity	Windy	Play Tennis
Sunny	Mild	High	False	No
Overcast	Hot	High	True	Yes
Rain	Cool	Normal	False	Yes
Sunny	Hot	Normal	True	Yes
Rain	Mild	High	True	 Yes
Overcast	Mild	High	False	Yes

$T_0: 1/6$

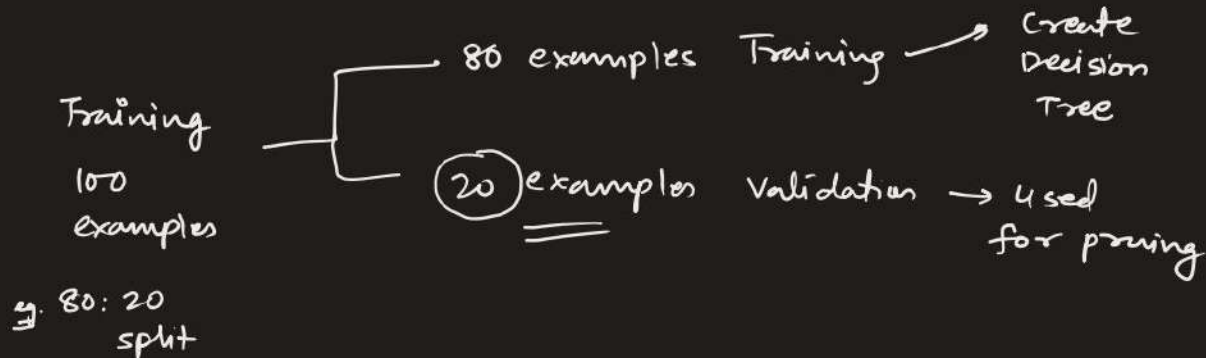
$T_1: 2/6$

$T_2: 0/6 \rightarrow$ min error
Free

$T_3: 1/6$

$T_4: 1/6$

pick the tree T_2 as it has min. misclassification rate on validation set.



Minimum Impurity Decrease:-

Instead of coming up with a information gain / Gini threshold while creating the decision tree, we can build the entire decision tree and then come up with a threshold for IG/Gini and then remove the nodes with IG less than the threshold in post-pruning stage.

Minimum Leaf Size:-

Remove leaf nodes with fewer samples than a specific threshold in post-pruning stage.